

WIRSCHING’S WEAK COVERING CONJECTURE: AN EXTENDED COMPUTATION, A PROVEN BOUND, AND THE STRUCTURE OF WHAT RESISTS IT

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ABSTRACT. Wirsching’s 1998 Weak Covering Conjecture asks for a sub-exponential $K(l)$ such that a large enough set $R_{j-1,j} \subset \mathbb{Z}_3^*$, built from strictly decreasing exponent tuples, is forced to cover every unit residue modulo 3^l . We extend the exact computation of $j^*(l)$, the smallest such covering budget, from $l = 20$ to $l = 23$, and compare four growth models for $e(l) := j^*(l) - l \log_4 3$ against the extended table: a constant model is disfavored at every statistic checked, but logarithmic, square-root, and slow-linear growth remain statistically indistinguishable from one another. We give the best bound known to us, $j^*(l) \leq \frac{9}{8}l + O(1)$, via a mean-payoff-game relaxation of the covering budget whose proof rests on one computationally verified, not generally proven, correspondence. We then examine the natural Fourier approach to a tighter bound: a uniform square-root-cancellation argument is capped below even this bound, a direct computation at one accessible level finds the bulk of the relevant exponential sum’s mass sitting outside any sparse exceptional set, and the residues that actually fail to be covered sit at extremal cells that a phase-randomization experiment shows are decided by phase structure that magnitude alone does not capture. We prove a theorem on the uncovered residues themselves: the set left uncovered at one budget is contained in the union of its double and its quadruple at the next, giving an explicit recursive upper bound on $j^*(l)$ that is empirically exact at every budget checked. We falsify the natural cost-1 form of a local repair rule this theorem might suggest, with explicit witness-level exhaustive counts, prove that every last-uncovered residue is $\equiv 1 \pmod{3}$, and report several further exact identities, verified but not proven, governing the last residues to be covered. Two precisely stated open questions remain, each narrower than the conjecture itself. The Weak Covering Conjecture remains open.

Keywords: Collatz map, 3-adic dynamics, covering conjecture, mean-payoff games, exponential sums, Littlewood–Offord theory.

2020 Mathematics Subject Classification: 11B50, 11K70, 91A50, 11L07, 37A44.

1. INTRODUCTION

Wirsching’s 1998 monograph [1] studies the density of predecessor sets under the $3n + 1$ map through an averaging construction on the 3-adic integers. A limiting Markov chain built from that construction has an invariant density, and Wirsching’s own route to positive predecessor density passes through a covering question about a family of finite subsets of $\mathbb{Z}_3^*$, the 3-adic units. For integers $j, k \geq 0$, write

$$(1) \quad R_{j,k} := \left\{ \sum_{i=0}^j 2^{a_i} 3^i \mid j+k \geq a_0 > a_1 > \cdots > a_j \geq 0 \right\} \subset \mathbb{Z}_3^*.$$

Each element of $R_{j,k}$ is built from a strictly decreasing sequence of $j + 1$ “digit positions” chosen from $\{0, \dots, j + k\}$: the i -th chosen position, counting from the top, contributes 2^{a_i} scaled by 3^i . Wirsching’s Weak Covering Conjecture (Conjecture 3.9, Chapter V, [1]) asks for a sub-exponential $K(l)$ such that $|R_{j-1,j}| \geq K(l) \cdot 3^l$ already forces $R_{j-1,j}$ to cover every unit residue modulo 3^l ; the cardinality on the left is that of the tuple family itself, before reduction, which by Proposition 1’s bijection equals $\binom{2j}{j}$, not the (a priori much smaller) size of its image modulo 3^l . Equivalently,

writing $j^*(l)$ for the smallest j such that $R_{j-1,j}$ covers $(\mathbb{Z}/3^l\mathbb{Z})^*$ exactly, the conjecture concerns the growth rate of

$$(2) \quad e(l) := j^*(l) - l \log_4 3,$$

the excess of the true covering budget over the leading-order rate a cardinality argument alone predicts ($|R_{j-1,j}| = \binom{2j}{j} \sim 4^j$, so matching $2 \cdot 3^{l-1}$ residues needs $j \sim l \log_4 3$ at leading order).

An earlier, unpublished manuscript of the author computed $j^*(l)$ exactly for $l = 1, \dots, 20$ by direct enumeration and reported the resulting table without a growth-rate analysis. This paper does three things. First, it extends that table to $l = 23$ and gives a proper statistical comparison of candidate growth models for $e(l)$ against the full range (Sections 2 and 3). Second, it gives the best upper bound on $j^*(l)$ we have found, via a reformulation of the covering budget as a mean-payoff game (Section 4). Third, and at greatest length, it reports a sustained attempt to close the gap to the conjectured rate directly, first by the natural Fourier approach, and, once that approach is shown to fail for a precise and checkable reason, by a direct study of the residues that actually resist coverage (Sections 5 and 6). That second attempt does not settle the conjecture. It produces a theorem on those residues, a falsification of the cost-1 form of a repair rule the theorem might suggest, and two precisely named open questions, each narrower than the conjecture itself and worth a dedicated attack in its own right.

2. THE COMPUTATION

Table 1 gives $j^*(l)$ and $e(l)$ for $l = 1, \dots, 23$. The values for $l \leq 20$ reproduce the earlier manuscript's table exactly; they were independently recomputed here by a from-scratch Rust reimplement of the covering search (a bitset rotation dynamic program over residues modulo 3^l), validated against the original Python implementation at every $l \leq 20$ before being trusted for the extension. The values for $l = 21, 22, 23$ are new.

l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$
1	1	0.208	9	13	5.868	17	21	7.528
2	4	2.415	10	15	7.075	18	22	7.735
3	6	3.623	11	16	7.283	19	23	7.943
4	7	3.830	12	17	7.490	20	24	8.150
5	9	5.038	13	18	7.698	21	25	8.358
6	10	5.245	14	19	7.905	22	26	8.565
7	11	5.453	15	20	8.113	23	27	8.773
8	12	5.660	16	20	7.320			

TABLE 1. $j^*(l)$ and $e(l) = j^*(l) - l \log_4 3$, $l = 1, \dots, 23$.

The single plateau in the table, $j^*(15) = j^*(16) = 20$, is not an error: $l = 15 \rightarrow 16$ is the only increment among the 22 steps from $l = 1$ to $l = 23$ at which the covering budget does not increase, and it is exactly the event Section 3's plateau-frequency test measures.

$l = 21$ fits in physical RAM once a memory-wasteful parallelization step in the original implementation was corrected. $l = 22$ required 500 GiB of swap, added to the machine specifically for this computation, and took approximately 10,749s (~ 3 hours) with swap I/O as the dominant cost, against 748s for $l = 21$ without swap. $l = 23$ required approximately 263 GiB of state and 375,615s (~ 104 hours); the successful run had no interruption, so a checkpoint/resume mechanism added after an earlier, unrelated policy-triggered kill of an interrupted attempt at the same level was never exercised on it. Three attempts at $l = 24$, made over the course of this project on the same machine, each failed before completion (two to memory-policy kills, one to an apparent out-of-memory condition with no surviving kernel log confirming it directly); $l = 24$ is not attempted further, and the table stops at $l = 23$.

3. THE GROWTH OF $e(l)$

We compare four candidate functional forms for $e(l)$'s growth against the tail $l = 10, \dots, 23$ ($n = 14$; the same range choice as the earlier manuscript, now with three more points), each fit by ordinary least squares: $e(l) = c$ (constant), $e(l) = a + b \ln l$ (logarithmic), $e(l) = a + b\sqrt{l}$ (square-root), and $e(l) = a + bl$ (slow-linear). Table 2 gives the fit statistics.

model	k	ΔAIC	ΔBIC	LOOCV RMSE
constant	1	16.091	15.452	0.5200
logarithmic	2	1.512	1.512	0.2991
square-root	2	0.722	0.722	0.2909
slow-linear	2	0.000	0.000	0.2837

TABLE 2. Model comparison on $e(l)$, $l = 10, \dots, 23$. $\Delta\text{AIC}/\text{BIC}$ relative to the best-fitting model (slow-linear).

A constant model is disfavored sharply and consistently: $\Delta\text{AIC} = 16.09$, $\Delta\text{BIC} = 15.45$, both growing at every l added since $l = 21$. The three growth models remain statistically indistinguishable from one another by the conventional $\Delta\text{AIC} > 2$ rule of thumb, with slow-linear taking a narrow lead on both AIC and leave-one-out cross-validation; the three candidate regressors ($\log l$, $\sqrt{l}$, l) are themselves highly correlated over this range (pairwise correlation ≥ 0.993), so this ranking says more about how thin the margin currently is than about which functional form is correct.

A direct combinatorial test tells the same story from a different angle. Among the 22 increments $j^*(l+1) - j^*(l)$ for $l = 1, \dots, 22$, exactly one is zero (the $l = 15 \rightarrow 16$ plateau noted above). The earlier manuscript's own table, $l \leq 20$, showed 3.9 plateaus expected in 19 increments under a constant- $e(l)$ rounding model; taking $q = 3.9/19 \approx 0.2053$ as that model's plateau probability and testing the observed count against $\text{Bin}(22, q)$, one-sided, gives $p = 0.0426$, crossing the conventional 0.05 threshold for the first time with the $l = 23$ point included; restricted to the same tail the model comparison above uses ($n = 13$), the same test gives $p \approx 0.22$. We report both figures: $e(l)$ is an exact, deterministic sequence, so neither AIC/BIC nor this test carries the sampling interpretation of a classical inference over noisy data. Both are best read descriptively, as "how well does a constant- $e(l)$ source fit the observed increment pattern."

A constant $e(l)$ is disfavored statistically by every measure above. It is also, unconditionally, impossible.

Proposition 1 (Unboundedness of $e(l)$). $e(l) \geq \frac{1}{4} \log_2 l - O(1)$.

Proof. Distinct exponent tuples give distinct sums: since the a_i are distinct, a_{j-1} is the unique minimal one, so $V = \sum_i 2^{a_i} 3^i$ has 2-adic valuation exactly a_{j-1} (every other term is divisible by a strictly higher power of 2), recovering a_{j-1} from V directly. The full term $2^{a_{j-1}} 3^{j-1}$ is then subtracted off, and the process iterates on the remaining $j-1$ terms. So $R_{j-1,j}$, in bijection with j -subsets of the $2j$ -element set $\{0, \dots, 2j-1\}$, has exact cardinality $\binom{2j}{j}$. Covering $(\mathbb{Z}/3^l\mathbb{Z})^*$, of size $2 \cdot 3^{l-1}$, requires the domain to be at least as large as the codomain: $\binom{2j}{j} \geq 2 \cdot 3^{l-1}$ is necessary for $j \geq j^*(l)$. The explicit bound $\binom{2n}{n} \leq 4^n / \sqrt{3n+1}$, valid for every $n \geq 1$, turns this into $4^j \geq 2 \cdot 3^{l-1} \sqrt{3j+1}$, hence

$$j \geq l \log_4 3 - \log_4 3 + \log_4 2 + \frac{1}{2} \log_4(3j+1).$$

Dropping the last, non-negative term already gives $j \geq l \log_4 3 - O(1)$, so $3j+1 = \Omega(l)$, hence $\log_4(3j+1) \geq \log_4 l - O(1) = \frac{1}{2} \log_2 l - O(1)$. Feeding this back into the displayed inequality gives the claimed bound. $\square$

4. A MEAN-PAYOFF-GAME UPPER BOUND

Reading an admissible tuple's exponents from the bottom up encodes covering as a game of imperfect information. A state is a unit $z \in \mathbb{Z}_3^*$. A move of cost $d \geq 0$ is *legal* at z when $2^d z \equiv 1 \pmod{3}$ (so that $T_d(z) := (2^{d+1}z - 2)/3$ is an integer), and *safe* at window k when, for $m := 3^k$, every one of the three residues $z, z+m, z+2m$ modulo $3m$ maps under T_d back into a unit modulo m . A minimizing player, seeing only $z \bmod 3^k$, chooses a legal, safe d ; an adversary then reveals the hidden next 3-adic digit $\epsilon \in \{0, 1, 2\}$, forcing the true state to $T_d(z + 3^k \epsilon) \bmod 3^k$. Both requirements matter: legality alone can send a unit to 0 (take $z = 1, d = 0$: $T_0(1) = 0$), and the safety filter is exactly what keeps the walk inside $\mathbb{Z}_3^*$ regardless of which digit the adversary supplies. Legality depends on d only through $d \bmod 2$ (since 2 has order 2 modulo 3), and 2 is a primitive root modulo 3^{k+1} for every $k \geq 0$ (order $2 \cdot 3^k$, by an elementary lifting-the-exponent induction), so $2^{d+1} \bmod 3^{k+1}$, and hence $T_d(z) \bmod 3^k$ and the safety check itself, depend on d only through $d \bmod (2 \cdot 3^k)$: successor states repeat with this period as d grows, and every representative beyond the smallest in its residue class has strictly larger cost for the same effect, so is never used by an optimal policy. The action set at each state is therefore, without loss, the subset of $\{0, \dots, 2 \cdot 3^k - 1\}$ that is legal and safe there, which is finite. That this subset is nonempty at every unit state has been verified computationally at every $k = 3, \dots, 14$ used below, not proven in general here.

This restricted-visibility game is, by construction, a two-player *mean-payoff game* on the $2 \cdot 3^{k-1}$ unit states modulo 3^k , with the now finite legal, safe moves as the minimizer's edges, the adversary's three choices of ϵ as the maximizer's edges, and payoff the long-run average of the move costs d . The classical theory of such games, initiated by Ehrenfeucht and Mycielski [2], guarantees a value and an optimal positional policy in general; Theorem 3 below needs none of that generality, only the twelve specific policies actually exhibited and directly checked in Table 3.

Empirical Result 2 (Soundness of the game). At full precision ($k = l$: the true state $z \bmod 3^l$ known exactly, no adversary), a play needs only legality, since z being a unit throughout makes safety automatic: each step chooses d with $2^d z \equiv 1 \pmod{3}$ and moves to $T_d(z)$. For every $l \geq 1$ and every unit z modulo 3^l , a legal l -step play with total cost J starting from z witnesses $z \in R_{J-1, J} \bmod 3^l$: reading the costs $d_0, \dots, d_{J-1}$ from the last step to the first recovers exactly the exponent gaps of an admissible tuple. Verified against direct brute-force enumeration of $R_{j-1, j}$'s covering sets for $l = 1, \dots, 6$, and against the known exact table of $j^*(l)$ (via the full-precision game's worst-case optimal cost) for $l = 1, \dots, 12$, with no discrepancy. We do not have a general proof.

A window- k policy is a legal full-precision play in this sense too: safety was defined precisely so that whichever of the three lifts $z, z+3^k, z+2 \cdot 3^k$ turns out to be the true state modulo 3^{k+1} , the move chosen from $z \bmod 3^k$ stays legal there, so running a window- k policy against the actual digit sequence gives a legal full-precision play, and Empirical Result 2 applies to it directly.

Theorem 3 (Window relaxation bound, conditional on Empirical Result 2). *For each $k = 3, \dots, 14$, the certified policy underlying Table 3 (a legal, safe move at every unit state modulo 3^k , average cost ρ_k) gives an explicit rational C_k with $j^*(l) \leq \rho_k \cdot l + C_k$ for all l .*

Proof. Fix k in range. The policy σ_k , the value ρ_k , and a potential $h_k : (\mathbb{Z}/3^k\mathbb{Z})^* \rightarrow \mathbb{Q}$ are computed by a nested Howard strategy-improvement solver and checked, state by state and digit by digit over the finite action set established above, against

$$d_{\sigma_k}(s) + h_k(s'(\epsilon)) \leq \rho_k + h_k(s)$$

at every state s and every adversary digit ϵ , where $s'(\epsilon)$ is the successor under σ_k 's chosen move; the matching adversary lower bound extracted from the same run certifies ρ_k exactly. Summing

the inequality along any l -step play $s_0, \dots, s_l$ telescopes to

$$\sum_{i=0}^{l-1} d_{\sigma_k}(s_i) \leq l\rho_k + h_k(s_0) - h_k(s_l) \leq l\rho_k + (\max h_k - \min h_k) =: l\rho_k + C_k,$$

regardless of which digits the adversary actually supplies, since the potential inequality holds for every ϵ at every step. σ_k needs only $z \bmod 3^k$ to choose each move, so it is executable at full precision too, giving a legal l -step play at z with total cost at most $l\rho_k + C_k$; by Empirical Result 2, this bounds the covering budget for z modulo 3^l . Taking the worst-case z gives $j^*(l) \leq \rho_k l + C_k$. $\square$

The gap between this and a fully unconditional theorem is narrow: only the *soundness* direction is needed (a completed game play constructs a witness), not the harder converse that the game's value equals $j^*(l)$ exactly.

ρ_k and C_k are computed exactly, as rationals, by a nested Howard strategy-improvement solver, self-certified at every k via a matching adversary lower bound extracted from the same run (upper bound equals lower bound, so the computed rational equals ρ_k exactly), and cross-validated by an independent value-iteration solver and, at $k = 4$ and $k = 7$, by running the resulting policy against the actual finite covering DP for every unit residue modulo 3^l , l up to 14, with zero illegal or unsafe moves encountered. Table 3 gives the values computed for $k = 3, \dots, 14$.

k	3	4	5	6	7	8	9	10	11	12	13	14
ρ_k	2	$\frac{5}{3}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{7}{5}$	$\frac{25}{19}$	$\frac{5}{4}$	$\frac{11}{9}$	$\frac{6}{5}$	$\frac{7}{6}$	$\frac{119}{104}$	$\frac{9}{8}$

TABLE 3. The window-relaxation value ρ_k , $k = 3, \dots, 14$.

At $k = 14$, $\rho_{14} = 9/8 = 1.125$, giving:

Corollary 4. $j^*(l) \leq \frac{9}{8}l + O(1)$.

This is, to our knowledge, the best bound on $j^*(l)$'s growth rate currently available, conditional on Empirical Result 2. ρ_k is observed to be non-increasing over the computed range $k = 3, \dots, 14$ (Table 3); we have not verified this in general, and Corollary 4 uses only the single value $k = 14$. The general technique, representing a covering-type combinatorial problem as a mean-payoff game and computing exact rational relaxation bounds this way, is not new to this application: beyond Ehrenfeucht and Mycielski's foundational theorem [2] itself, a close and recent methodological precedent proves rationality and computability of the covering radius of a primitive sofic shift by essentially the same construction, applied to a different object [3]. We cite both as precedent for the framing; what is new here is its specific application to $j^*(l)$.

Remark 5. Fitting $\rho_k = L + A/k$ by ordinary least squares to $k = 10, \dots, 14$ gives $L \approx 0.876$, still well above the conjectured true rate $\log_4 3 = 0.7925\dots$; the fit is sensitive to the range used (all of $k = 3, \dots, 14$ gives $L \approx 0.904$) and is not evidence of any specific limit. Only the direction $j^*(l) \leq \rho_k l + C_k$, proven above, is established. The reverse inequality, $\inf_k \rho_k \leq \limsup_l j^*(l)/l$, which would be needed to conclude anything about where ρ_k is heading, is not established, and there is evidence in the course of this work that the window relaxation may carry an irreducible gap above the true rate.

5. THE EXPONENTIAL SUM, AND WHY IT RESISTS A DIRECT ATTACK

Corollary 4's bound, $1.125l$, is well above the conjectured rate $\log_4(3)l \approx 0.7925l$. The natural route to closing that gap is Fourier-analytic. By orthogonality on $\mathbb{Z}/3^l\mathbb{Z}$, the number of tuples landing in residue class z is

$$(3) \quad N(z) = \frac{1}{3^l} \sum_{t=0}^{3^l-1} S(t) e\left(-\frac{tz}{3^l}\right), \quad S(t) := \sum_{j+k \geq a_0 > \dots > a_j \geq 0} e\left(\frac{t}{3^l} \sum_{i=0}^j 2^{a_i} 3^i\right),$$

and, since $S(0) = |R_{j,k}| =: T$, the triangle inequality gives $N(z) > 0$ for every z as soon as $\sum_{t \neq 0} |S(t)| < T$; a uniform bound $|S(t)| = o(T/3^l)$ at every nonzero t would suffice, at a rate controlled directly by the exponent in the bound. This approach does not work, for three separate and mutually reinforcing reasons.

5.1. A hard ceiling on uniform cancellation.

Proposition 6. *Any proof of covering via a uniform bound $|S(t)| \leq C\sqrt{T}$ for a fixed constant C (square-root cancellation), applied via the triangle inequality across all $3^l - 1$ nonzero frequencies, requires $j \geq 2 \log_4(3)l - O(\log l) \approx 1.585l - O(\log l)$.*

Proof. Write $T := |R_{j-1,j}| = \binom{2j}{j}$. Positivity of $N(z)$ for every z via $\sum_{t \neq 0} |S(t)| < T$ combined with the uniform estimate $|S(t)| \leq C\sqrt{T}$ over the $3^l - 1$ nonzero frequencies forces $(3^l - 1)C\sqrt{T} < T$, so $T > ((3^l - 1)C)^2$, i.e. $4^j / \sqrt{3j + 1} \geq T > ((3^l - 1)C)^2$ (using the explicit bound from Proposition 1's proof), which gives $j \geq 2 \log_4(3)l - O(1)$. $\square$

Proposition 6's threshold is *worse* than Corollary 4's bound: any approach that only delivers generic square-root cancellation across every frequency cannot even match what is already known by other means, let alone improve on it. This rules out the naive version of the Fourier approach cleanly, and directs attention to whether a finer, frequency-dependent treatment (a major/minor-arc split, in the language of the circle method) can do better.

5.2. A computation: the exceptional mass does not stay small. Numerically, $|S(t)|$ is far from uniform: the frequencies of largest magnitude sit close to dyadic rationals of small denominator, $t/3^l \approx p/2^r$ for small r . The natural repair for Proposition 6's ceiling is to excise these major-arc resonances and control the remainder by an averaged bound; whether that remainder stays sparse enough at the scale that matters is a question this section answers computationally, at $l = 18$, with a negative result. A fixed threshold makes the visible spikes look sparse; the threshold $\eta \sim c/\sqrt{T}$ that actually governs the L^1 norm is different and gives a different picture: summing every one of the 8,014 coefficients $S(t)$ with t coprime to 3 that exceed a generous fixed threshold of $0.001 \cdot T$ contributes less than 132 to an L^1 norm of 5226, so over 97% of the mass at this single computed level lives in an exponentially numerous population of near-average-magnitude coefficients. This is evidence against a sparse-exceptional-set repair at accessible l ; it says nothing about every l .

The same clustering near dyadic rationals connects the problem to a 2011 discussion of Terence Tao's, applying Littlewood–Offord and Riesz-product methods to an object built from the same $2^a 3^b$ -type exponential sums, in the closely related setting of a hypothetical non-trivial Collatz cycle [4]; his construction is structurally close to a boundary slice of $R_{j,k}$ with the bottom exponent pinned at 0. His own conclusion, in his own words, is that “once one uses the transcendence theory bound, ... Littlewood–Offord techniques are no longer of much use,” and that he knows “of no way to get a nontrivial separation property between powers of 2 and powers of 3 other than via transcendence theory methods.”

5.3. A direct test of the residues that actually fail. The most direct test examines the residues that actually fail to be covered, one budget step before covering succeeds. At every level checked, their count is tiny (single or low double digits) against an ambient occupancy mean in the hundreds, sitting in cells whose local intensity, at every resolution checked, would assign them probability below e^{-20} under a locally homogeneous model. Holding every $|S(t)|$ fixed while randomizing the phases of conjugate-symmetric pairs collapses the observed extremal statistics to what a generic-phase model predicts; the actual, unscrambled phases do not. At the levels checked, coverage failure is decided by phase structure that magnitude-only methods (uniform bounds, averages, low moments of $S(t)$) do not see.

Proposition 7. $|S(3^{l-1})|/\binom{2m}{m} \rightarrow 1/\sqrt{3}$ as $m \rightarrow \infty$, for the family $R_{m-1,m}$, for every fixed $l \geq 1$.

Proof sketch. $\sum_i 2^{a_i} 3^i \equiv 2^{a_0} \pmod{3}$, so $S(3^{l-1})/\binom{2m}{m}$ depends only on the limiting distribution of the top exponent a_0 's parity as $m \rightarrow \infty$, which converges to the mixture $\frac{1}{3}\omega + \frac{2}{3}\omega^2$ of the two nontrivial cube roots of unity $\omega = e^{2\pi i/3}$ (the limiting gap $2m-1-a_0$ of the maximum of a random m -subset of $\{0, \dots, 2m-1\}$ is geometric with parameter $1/2$, giving $\Pr[a_0 \text{ even}] \rightarrow 1/3$). This mixture equals $-\frac{1}{2} - \frac{i\sqrt{3}}{6}$, of modulus $1/\sqrt{3}$. $\square$

Uniform decay across every nonzero frequency, the premise a naive statement of the Fourier approach might suggest, is false at this explicit family of frequencies.

6. A THEOREM ON THE RESIDUES THAT RESIST COVERAGE

Write $H(l, j)$ for the set of units modulo 3^l not covered by $R_{j-1,j}$, so that $j \geq j^*(l)$ exactly when $H(l, j) = \emptyset$. Every result in this section needs $j \geq l$; at every computed level $j^*(l) > l$ (Table 1), but the conjectured rate $\log_4 3 < 1$ would eventually put $j^*(l)$ below l , outside this section's reach. Within this range, emptiness persists: Theorem 8 below gives $H(l, j) = \emptyset \Rightarrow H(l, j+1) \subseteq 2\emptyset \cap 4\emptyset = \emptyset$, so “ $H(l, j) = \emptyset$ exactly when $j \geq j^*(l)$ ” is consistent for every $j \geq l$. We make no claim about $j < l$.

Theorem 8 (Holdout doubling inclusion). *For $j \geq l$, $H(l, j+1) \subseteq 2H(l, j) \cap 4H(l, j)$.*

Proof. $R_{j-1,j}$ is the set of values $r = \sum_{i=0}^{j-1} 2^{b_i} 3^i$ over j -subsets $\{b_0 > \dots > b_{j-1}\} \subseteq \{0, \dots, 2j-1\}$, and $R_{j,j+1}$ the same with $(j+1)$ -subsets of $\{0, \dots, 2j+1\}$. Fix $r \in R_{j-1,j}$ with exponent set S , and $s \in \{1, 2\}$. Shifting every exponent up by s gives $S+s \subseteq \{s, \dots, 2j-1+s\} \subseteq \{1, \dots, 2j+1\}$, a j -subset whose induced value is $2^s r$ (a uniform shift of exponents preserves relative rank, so each term's power of 3 is unchanged). Since $S+s$ contains no 0, adjoining 0 as the new smallest exponent gives a $(j+1)$ -subset of $\{0, \dots, 2j+1\}$: it becomes the new rank- j term, contributing $2^0 \cdot 3^j = 3^j$. So this witness represents $2^s r + 3^j$, and $3^j \equiv 0 \pmod{3^l}$ once $j \geq l$. Hence $2R_{j-1,j} \cup 4R_{j-1,j} \subseteq R_{j,j+1} \pmod{3^l}$ for $j \geq l$. So if $z \in H(l, j+1)$, i.e. z is not represented at budget $j+1$, then z lies in neither $2R_{j-1,j}$ nor $4R_{j-1,j}$: both $z/2$ and $z/4$ are unrepresented at budget j , i.e. $z \in 2H(l, j) \cap 4H(l, j)$. $\square$

Corollary 9 (Chain contraction). *If $x, 2x, \dots, 2^{t-1}x \in H(l, j+1)$ (all reduced modulo 3^l , $t \geq 1$), then $x/4, x/2, x, \dots, 2^{t-2}x \in H(l, j)$, a chain of length $t+1$.*

Proof. Apply Theorem 8 to each of the t elements: $2^i x \in H(l, j+1)$ gives $2^{i-1}x, 2^{i-2}x \in H(l, j)$ for $i = 0, \dots, t-1$. The union of these pairs over all i is exactly $\{x/4, x/2, x, \dots, 2^{t-2}x\}$, a doubling chain of length $t+1$. $\square$

Corollary 10 (Run-length bootstrap). *Writing $\text{maxrun}(H)$ for the greatest t such that H contains t pairwise distinct elements of the form $x, 2x, \dots, 2^{t-1}x$ (all reduced modulo 3^l), with $\text{maxrun}(\emptyset) := 0$, $j^*(l) \leq j + \text{maxrun}(H(l, j))$ for every $j \geq l$.*

Proof. By Corollary 9, $\text{maxrun}(H(l, j)) \geq \text{maxrun}(H(l, j+1)) + 1$ whenever $H(l, j+1)$ is nonempty, so $\text{maxrun}(H(l, j))$ strictly increases by at least 1 as j decreases from $j^*(l)$; equivalently it strictly decreases by at least 1 at every budget step increasing from l , until it reaches 0, at which point H is empty. $\square$

Empirical Result 11. Corollary 10 is empirically exact, $j^*(l) = j + \text{maxrun}(H(l, j))$, at every (l, j) with $l+1 \leq j \leq j^*(l)$ we were able to compute, $l = 5, \dots, 20$ at every such budget and l up to 22 at the single budget $j = l+1$, including a case with $|H(l, j)| > 3 \times 10^6$. (Past $j^*(l)$, $H(l, j) = \emptyset$ and $\text{maxrun}(\emptyset) = 0$, so the displayed equality no longer holds there; the claim is only about budgets up to and including $j^*(l)$.)

The converse direction, that a doubling chain of the observed maximal length is not merely sufficient but necessary for survival to that budget, is not proven, and is false as an unrestricted statement: $1547 \in 2H(7, 8) \cap 4H(7, 8)$, yet $1547 \notin H(7, 9)$ (both of 1547's two budget-9 witnesses span the full exponent range, the one case Theorem 8's argument does not directly control). We return to a restricted, still-open form of this converse in Section 7.

A natural strengthening of Theorem 8, that every individual witness admits a bounded-cost local repair, is also false in its most optimistic form.

Call the *Durfee depth* of a witness $\{a_0 > \dots > a_{j-1}\} \in R_{j-1,j}$ at budget j the count $\#\{i : a_i \geq j\}$, the number of chosen exponents at or above the budget itself.

Empirical Result 12 (Failure of the cost-1 local repair rule). At the budget transition $(l, j) = (6, 10) \rightarrow (7, 11)$, 100 of the 1458 children (residues newly required for coverage at the larger budget) have minimum achievable Durfee depth exceeding their parent's minimum depth by more than one, up to three; at $(7, 11) \rightarrow (8, 12)$, 332 of 4374 children do, likewise up to three. A correspondingly exhaustive check, measuring witness repair cost as the minimum number of exponents that must change between a child's cheapest witness and its cheapest parent witness (matched by the natural containment of exponent sets), finds distances up to 7 at both transitions, against a distance of 1 for the overwhelming majority. This is evidence against a repair cost of exactly 1 at these two transitions; it does not establish that no bounded repair cost exists.

This falsifies, in its originally proposed cost-1 form, a natural conjecture this project had earlier entertained (that coverage propagates via a uniform, cost-1 local repair rule budget by budget); its valid surviving content is Theorem 8 and its corollaries above.

7. FURTHER STRUCTURE OF THE LAST HOLDOUTS

The residues in $H(l, j^*(l) - 1)$, the last to be covered, satisfy several further exact laws. For $W \geq l - 1$, write

$$U(l, W) := \left\{ \sum_{i=0}^{l-1} 2^{\beta_i} 3^i \pmod{3^l} \mid W \geq \beta_0 > \dots > \beta_{l-1} \geq 0 \right\},$$

an l -term analogue of $R_{j-1,j}$ built from a fixed term count and a growing exponent ceiling W .

Lemma 13 (Width scaling identity). *For $j \geq l \geq 1$,*

$$(4) \quad R_{j-1,j} \equiv 2^{j-l} U(l, j+l-1) \pmod{3^l}.$$

Proof. Since $3^i \equiv 0 \pmod{3^l}$ for $i \geq l$, the residue of $r = \sum_{i=0}^{j-1} 2^{b_i} 3^i \in R_{j-1,j}$ modulo 3^l depends only on its top l exponents $b_0 > \dots > b_{l-1}$. A decreasing l -tuple $b_0 > \dots > b_{l-1} \geq 0$ with $b_0 \leq 2j-1$ extends to a full admissible j -tuple in $\{0, \dots, 2j-1\}$ (appending $j-l$ further, strictly smaller, distinct exponents) exactly when $b_{l-1} \geq j-l$: append $b_{l-1}-1, \dots, b_{l-1}-(j-l)$ when this holds, and no extension exists when it does not, since fewer than $j-l$ nonnegative integers lie below b_{l-1} . Writing $c_i := b_i - (j-l)$ turns $b_{l-1} \geq j-l$ into $c_{l-1} \geq 0$ and $b_0 \leq 2j-1$ into $c_0 \leq 2j-1-(j-l) = j+l-1$, and $\sum_i 2^{b_i} 3^i = 2^{j-l} \sum_i 2^{c_i} 3^i$. So the top- l -exponent tuples realizable by $R_{j-1,j}$ are exactly 2^{j-l} times the decreasing l -tuples counted by $U(l, j+l-1)$, giving the claimed identity modulo 3^l . $\square$

In particular $j^*(l)$ is the smallest $j \geq l$ with $U(l, j+l-1) = (\mathbb{Z}/3^l\mathbb{Z})^*$, and writing $W^*(l) := j^*(l) + l - 2$ for the width one step before that,

$$(5) \quad H(l, j) = 2^{j-l} K(l, j+l-1), \quad K(l, W) := (\mathbb{Z}/3^l\mathbb{Z})^* \setminus U(l, W).$$

Theorem 14 (Last-holdout parity). *If $j^*(l) \geq l+1$, every element of $H(l, j^*(l)-1)$ is $\equiv 1 \pmod{3}$. This covers $l = 2, \dots, 23$ (Table 1); it fails only at $l = 1$ ($j^*(1) = 1$, outside (5)'s range), and would fail again for any l at which the conjectured rate $\log_4 3 < 1$ eventually pushes $j^*(l)$ below $l+1$.*

Proof. $U(l, \cdot)$ is monotone increasing in W , so $K(l, \cdot)$ is monotone decreasing; every $x \in K(l, W^*(l))$ is, by minimality of $j^*(l)$, representable at width $W^*(l) + 1$ but not at $W^*(l)$. Any witness for x at width $W^*(l) + 1$ using only exponents $\leq W^*(l)$ would already witness x at that smaller width, so every such witness has top exponent $\beta_0 = W^*(l) + 1$. Its contribution to the value, at rank 0, is $2^{\beta_0} \cdot 3^0 = 2^{W^*(l)+1}$, so $x \equiv 2^{W^*(l)+1} \equiv (-1)^{W^*(l)+1} \pmod{3}$. By (5), which needs $j^*(l) - 1 \geq l$, elements of $H(l, j^*(l) - 1)$ are $2^{j^*(l)-1-l}x$ for such x , hence

$$2^{j^*(l)-1-l}x \equiv (-1)^{j^*(l)-1-l} \cdot (-1)^{W^*(l)+1} = (-1)^{2j^*(l)-2} = 1 \pmod{3},$$

using $W^*(l) + 1 = j^*(l) + l - 1$ and that $2j^*(l) - 2$ is always even. $\square$

Empirical Result 15 (Near-extinction bijection). $H(l, j^*(l) - 1) = 2 \cdot \{x \in H(l, j^*(l) - 2) : x \equiv 2 \pmod{3}\}$, exactly, verified at every $l = 2, \dots, 9$ against the exact table with no exception.

The same width mechanism that proves Theorem 14 governs this bijection one step earlier, through the second-highest exponent β_1 of the extremal witness; unlike the top-exponent argument above, pinning β_1 down needs the one-step width recursion $U(l, W + 1) = U(l, W) \cup 2U(l, W) \cup \text{Corner}(l, W + 1)$ stated below; we omit the full derivation and record the identity as verified.

An earlier stage of this investigation reported a cross-level “ $\times 4$ -only” pattern relating $H(l + 1, j^*(l + 1) - 1)$ to $H(l, j^*(l) - 1)$. Writing π_l for reduction modulo 3^l , the precise congruence this suggests, $x \equiv 4\pi_l(x) \pmod{3^{l+1}}$, is impossible whenever $\pi_l(x)$ is a unit and $l \geq 2$: reducing both sides modulo 3^l forces $\pi_l(x) \equiv 4\pi_l(x) \pmod{3^l}$, i.e. $3\pi_l(x) \equiv 0 \pmod{3^l}$. Checked exhaustively at $l = 2, \dots, 6$, $\pi_l(x)$ never lies in $H(l, j^*(l) - 1)$ for $x \in H(l + 1, j^*(l + 1) - 1)$ in the first place, so the pattern has no witnessing instance at these levels either. No cross-level relation between $H(l + 1, \cdot)$ and $H(l, \cdot)$ is established by this paper.

Empirical Result 16 (Mod-9 two-class bound). Writing $J := j^*(l)$, at every $l = 3, \dots, 9$ checked, $H(l, J - 1) \pmod{9} \subseteq \{4^J, 4^{J+1}\} \pmod{9}$, and the class actually occupied is 4^{J+1} , never 4^J , at every level checked. The two-class containment follows the same mechanism as Theorem 14 one digit further (the mod-9 class of a threshold witness is $2^{\beta_0} + 3 \cdot 2^{\beta_1} \pmod{9}$, with β_0 pinned and β_1 's parity giving the two options); excluding 4^J specifically is the sharpest concrete open question this part of the investigation produced.

Proposition 17 (Sufficiency of bounded maxrun). *If $\text{maxrun}(H(l, l + 1))$ is bounded by a constant C for all l , then $j^*(l) \leq l + C + 1$.*

Proof. Corollary 10 at $j = l + 1$. $\square$

The converse is not established, and the natural argument for it does not work: reading Corollary 10 downward gives only $\text{maxrun}(H(l, l + 1)) \geq j^*(l) - l - 1$, a *lower* bound on maxrun, which does not force maxrun to stay bounded when $j^*(l) - l$ does. Proving the converse would need an upper bound on maxrun in terms of $j^*(l) - l$, equivalent to a tightness statement for Corollary 10 that Section 8's corner-redundancy question addresses and that is false in the strong, per-witness form: recall 1547 from Section 6.

A naive expectation, that a random set of holdouts at the observed density would have doubling chains far longer than the flat ceiling of 3–4 measured at every level $l = 5, \dots, 22$, does not survive contact with the actual, rapidly declining density of $H(l, l + 1)$. An independence model matched to that actual density tracks the observed sequence within about one unit at every level, including both the one rise and the one fall in it: $\text{maxrun}(H(l, l + 1))$ is typical for the set's density under that model. This does not, by itself, say anything about whether $\text{maxrun}(H(l, l + 1))$ is bounded; Proposition 17 is the only established link between that question and $j^*(l)$'s rate.

The one-step width identity is exact:

$$U(l, W + 1) = U(l, W) \cup 2U(l, W) \cup \text{Corner}(l, W + 1),$$

where $\text{Corner}(l, W + 1)$ consists of the witnesses using both the top exponent $W + 1$ and the bottom exponent 0 simultaneously. At every width $W \geq 2l + 1$ checked exhaustively, $l = 3, \dots, 13$, $\text{Corner}(l, W + 1)$ contributes nothing new: every corner witness's value is already covered without it. If this *corner-redundancy* property holds at every sufficiently large width in general, it would make $j^*(l)$ computable from a single holdout set, a converse to Corollary 10 and a real practical gain for extending Table 1 beyond $l = 23$; it remains unproven, with the only known failures at the boundary width $W = 2l$.

8. DISCUSSION

Two named questions summarize what stands between this paper's results and a sharper theorem. Deciding which of the two classes in Empirical Result 16 is excluded would turn the observed single-class pattern into a proven statement. Proving corner-redundancy at every sufficiently large width would make Corollary 10's bootstrap tight in general, not merely empirically exact at every computed instance, and would settle the converse to Proposition 17 left open above. Neither question is, on its face, as hard as the Weak Covering Conjecture itself; whether either is substantially easier is not known.

A finite exponential sum built from strictly decreasing, exponentially weighted digit tuples is a natural object beyond this specific covering problem: finite-population L -statistics at non-CLT frequencies and weights, Littlewood–Offord theory for structured coefficient sequences, and the analytic side of the Collatz literature via Tao's discussion are all adjacent areas. Three things found here may recur there under different names: generic cancellation capping out below what combinatorial counting already proves, exceptional mass that does not concentrate on a sparse set at the accessible scale, and coverage failure decided by an extremal arithmetic obstruction that a phase-randomization experiment showed magnitude-only methods miss.

The Weak Covering Conjecture itself, and the exact asymptotic rate of $e(l)$, remain open.

9. CODE AND DATA AVAILABILITY

The repository <https://github.com/faculdade/weak-covering-conjecture> is reserved for this paper's code and data, organized by section, and will be populated before submission.

Acknowledgments. AI assistance was used for textual review and translation.

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